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Intelligent Healthcare Management: An integrated system combining AI-powered appointment scheduling and medicine management.

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Abstract - As the healthcare services become more complex, intelligent systems are required for scheduling appointments, medication management and continuous care of patients. Patients often have difficulty getting to appropriate health care appointments, managing complicated medication regimens, following prescribed dosage schedules, and maintaining adherence to medications. The proposed Intelligent Healthcare Management system is based on the appointment scheduling with AI and the complete medicine management on a single platform. The appointment module allows patients to search for appropriate healthcare services and make appointments efficiently. The medicine management module allows the digitization of prescriptions using Optical Character Recognition (OCR), structured medicine management, dosage scheduling, medication reminders, drug-interaction detection, adherence monitoring and refill prediction. The system connects the appointment and medication workflows, so that the information gathered during the healthcare consultation can guide the next medication management and follow-up care. The system is proposed to reduce manual effort, improve medication organization, support timely appointments and provide a central platform for customized and continuous healthcare management.

Keywords: Intelligent Healthcare Management, AI-Powered Appointment Scheduling, Medicine Management, Optical Character Recognition, Prescription Digitization, Medication Adherence, Drug Interaction Detection, Refill Prediction.

I. INTRODUCTION

The healthcare services include a number of activities such as appointment scheduling, prescription management, medication schedule, and patient adherence monitoring. But often, these activities are handled through separate processes, making it difficult for patients to manage their healthcare

information efficiently. Patients may have difficulty obtaining suitable appointments, interpreting handwritten prescriptions, recalling medication schedules, and adhering to medication regimens. Such challenges can lead to missed appointments, inappropriate use of medications and diminished continuity of care.

Medication management is of particular importance for patients taking several medications or having complex medication taking schedules. Manual interpretation and entry of medicine name, dosage, frequency and instructions is common for handwritten or image based traditional prescriptions. This can be a long drawn out process and may lead to errors being introduced at the data entry stage. Similarly, standard appointment systems are limited to making appointments and availability, with no direct link between the consultation process and management of medications thereafter.

Current health care applications offer solutions to individual problems such as prescription management, appointment scheduling, medication reminders or adherence monitoring. However, there is a need for an integrated platform that combines these functions and enables the seamless flow of information from appointment scheduling to prescription processing and medication management. An intelligent system can help patients to organize their healthcare activities so as to reduce manual effort and improve access to related medical information.

The proposed Intelligent Healthcare Management system addresses these challenges by providing a single platform for AI-powered appointment scheduling and medicine management. The system leverages OCR-based prescription digitization to derive medicine-related data from images of prescriptions and convert them into structured medication records. It also provides medication scheduling, automated reminders, drug-interaction detection, adherence tracking, and refill prediction.

The appointment module facilitates efficient management of appointments and links the workflow of the consultation with the medication records of the patient.

The proposed system aims to provide a centralized and patient-centric healthcare platform that organizes the appointments and medicines in a better way and supports the continuous care. The system aims to reduce manual processes and improve medication adherence by incorporating appointment scheduling with prescription and medication management, providing a more efficient and personalized healthcare experience.

II. LITERATURE REVIEW

Intelligent Patient Appointment Schedules (Elhag et al., 2026) describes an intelligent approach to automate patient appointment scheduling. The system integrates Business Process Analysis, Bizagi workflow modeling, Six Sigma DMAIC and machine learning techniques. We use BiLSTM on BioBERT for symptom analysis and specialist matching, followed by automatic booking. The proposed system reported a simulated reduction of 96.3% in booking time and 70% in human intervention, showing the potential of AI based appointment scheduling to boost healthcare efficiency. However, the work is primarily centered on appointment scheduling and does not include prescription digitization or long-term medication management.

Ruiz Mejia and Rawat (2025) proposed “MedScrubCrew: A Medical Multi-Agent Framework for Automating Appointment Scheduling Based on Patient-Provider Profile Resource Matching”. The framework is based on a multi-agent architecture that combines Gale-Shapley stable matching, knowledge graphs and LLM-based agents to perform patient-provider matching and triage. This study shows how intelligent agents can automate appointment setting based on patient and provider profiles. The framework enhances contextual matching but is mainly focused on appointment allocation rather than linking the appointment information with prescription and medication management.

Wood et al. (2025) “Artificial intelligence machine learning-driven outpatient appointment management: A qualitative study on acceptability” The study on

acceptability of artificial intelligence and machine learning-based appointment management among healthcare staff and patients. User perceptions were identified via semi-structured interviews and thematic analysis. Users valued convenience, accuracy and reminders and considered privacy concerns, limited interaction and fragmented integration to be important barriers. This demonstrates that an intelligent healthcare system should not only automate scheduling but deliver an integrated and user-oriented experience.

North et al. (2025) “Improving Patient Appointment Scheduling: Results of an Automated Waitlist Process to Reduce Patient Wait Times for Appointments. The study used an automated waitlist and self-reschedule process to identify eligible patients and provide them with earlier appointment times. The system produced more than one million appointment offers for some 230,000 appointments, showing the scalability of automated waitlist management. But the work is focused on improving the availability of appointments and the waiting times instead of integrating the appointment scheduling with the patient medication workflows.

The paper “An adaptive decision support system for outpatient appointment scheduling with heterogeneous service times” (Feng et al., 2024) proposed a Cluster-Predict-Schedule (CPS) approach, which integrates unsupervised clustering, decision tree modeling, supervised machine learning, and Monte Carlo simulation. When creating appointment schedules, consideration is given to differences in service times. The study reported cost reductions of up to 15 percent over first-call-first-appointment scheduling and more than 4.7 percent savings over traditional sequencing approaches. The study emphasizes the benefit of smart scheduling and resource optimization; however, prescription processing and medication adherence are not covered.

Ala and Chen (2022) published “Appointment Scheduling Problem in Complexity Systems of the Healthcare Services: A Comprehensive Review” which reviewed more than 150 studies of healthcare appointment scheduling. This review grouped factors such as patient flow, patient preferences, arrival uncertainty, service-time uncertainty and scheduling approaches. Simulation, mathematical optimization, Markov models and artificial intelligence were identified as important approaches to reduce patient

waiting time and improve the utilization of healthcare resources. The review recognizes appointment scheduling as a challenging optimization problem in healthcare and substantiates the necessity for intelligent scheduling mechanisms.

Boone et al. (2022) evaluated an automated SMS reminder system using longitudinal clinic data in “How scheduling systems with automated appointment reminders improve health clinic efficiency”. Automated reminders improved timely cancelations and rescheduling and increased the capacity of clinics to serve additional patients, the study found. This shows the operational advantage of automated communication in appointment management. The system, however, is primarily geared toward appointment reminders and lacks an integrated medication-management workflow.

Ponnuru et al. (2024) proposed the “Image-Based Extraction of Prescription Information using OCR-Tesseract.” for prescription digitization. The method uses image pre-processing techniques such as resizing and adaptive thresholding, followed by OCR/Tesseract to extract medicine names, dosage information and patient instructions. The study reported about 98% accuracy and showed that OCR can reduce manual prescription data-entry effort and potential errors. But the work is mostly concerned with extracting prescription information and does not expand the extracted data to a complete medication and appointment management platform.

Datta et al. (2025) have proposed “Preserving medical information from doctor’s prescription ensuring relation among the terminology”. The system comprises YOLO-based Region of Interest detection, OCR-based extraction of prescription, spell correction and structured EHR/database integration. The study achieved 99.6% accuracy in detecting ROI and 96% accuracy in correcting misspellings of medication names, while preserving the relationships between the names of medicines, dosage, instructions and manufacturer information. This work is particularly relevant to the proposed system as it shows the importance of converting prescription images into useful and structured medical information. However, it is not as integrated as our system where appointments, medication adherence, reminders, and refill management are included.

Development of a new optical character recognition

system to reduce recording time for vital signs and prescriptions: The study “A simulation-based study (2023) discussed the use of machine learning-based OCR and pattern recognition for the reduction of manual recording effort. The OCR-based approach reduced the recording time to 18 seconds compared to 144 seconds required for manual entry for prescriptions with six drugs and generated fewer errors. This illustrates the practical benefit of digitizing prescriptions in terms of saving documentation time and manual workload.

Tabassum et al. (2022) suggested “An online cursive handwritten medical words recognition system for busy doctors in developing countries for ensuring efficient healthcare service delivery”. The researchers constructed a medical handwriting corpus and applied a Bidirectional LSTM with Rotate-Shift-Stretch (RSS) data augmentation for recognizing handwritten medical terms. The system achieved an average accuracy of 93.0% with a maximum of 94.5% which demonstrates the feasibility of handwritten medical terminology recognition. However, handwritten medical recognition is still a challenge and this paper mainly focuses on recognition, not complete prescription to medication management.

Anjani Kumar et al. (2022) have also developed “OCR based medical prescription and report analyzer”. The system employs image processing, handwriting character recognition and printed character recognition to extract information from medical prescriptions and reports. Its goal was to transform medical documents into structured information and to make it easier for patients to understand prescriptions and reports. The study, however, does not extend the extracted information to integrated medication scheduling, adherence monitoring or appointment management.

The reviewed studies show a significant progress in AI-based appointment scheduling and OCR-based prescription digitization, but these healthcare processes are mostly addressed separately, and continuous medication support is not provided. To address this gap, the Intelligent Healthcare Management system proposed in this paper combines AI-facilitated appointment scheduling, OCR-based digitization of prescriptions, medicine management, dosage scheduling, reminders, detection of drug-interactions, adherence monitoring, and refill

prediction into a singular platform, thereby connecting appointment booking with post-consultation medication management and fostering a more efficient, ongoing, and patient-centric healthcare experience.

III. METHODOLOGY

The proposed Intelligent Healthcare Management is an integrated system combining the features of appointment scheduling and prescription plus medicine management. The system comprises various interconnected modules implementing the complete workflow of the healthcare management, starting from patient appointment scheduling to prescription digitisation, medicine management, reminders, adherence-monitoring, safety-checking, and refill prediction. The multiple modules are connected through a central backend and relational database that ensures consistent maintenance of patient, appointment, prescription, and medication information.



Figure 3.1. Architecture Diagram of the system

The proposed system comprises a React frontend and a FastAPI backend with PostgreSQL as the primary database that stores information about users, appointments, prescriptions, medicines, and adherence. The frontend and backend communicate over REST API to enable the modules to interact with each other. Generally, the application is designed to support appointment scheduling, prescription scanning and recognition, medicine organization, reminders, safety monitoring, adherence tracking, and refill prediction in a single healthcare platform. Firstly, the system allows scanning of the patient's script, converting the information into digital form and creating a medication

record in the backend. Prescriptions can also be manually entered or uploaded for processing and generating appropriate medical data records. Medication records are subsequently utilized to schedule reminders, conduct safety permission checks, adherence tracking, refill prediction, and appointment scheduling. In addition, the caregiver module allows caregivers to monitor the medication process of their managed patients. Therefore, the overall system architecture is focused on linking appointment scheduling and medication management while addressing patient medication activities as shown in the figure below.

APPOINTMENT MANAGEMENT MODULE

The appointment module handles the complete procedure of appointment scheduling between patients and healthcare providers. The patients can browse through the healthcare services, choose desired appointment slots, and submit appointment requests. In turn, the system uses intelligent scheduling mechanisms to consider available slots and patient and provider information during allocation. Appointment records are subsequently stored through the central database, enabling the users and administrators to monitor scheduled, completed, or cancelled appointments. Generally, the module presents an initial stage of the healthcare workflow and provides the information for subsequent consultation and medication management procedures.

PRESCRIPTION OCR MODULE

The prescription digitisation module presents the OCR component that transforms the uploaded prescription images into medical information. The uploaded image undergoes processing steps to enhance the quality of the input before extracting text. The extracted information is analysed to identify relevant medicine details such as the medicine name, dosage, frequency, and instructions. The information is subsequently formatted into structured medication records, reducing the need for manual entry and establishing a direct link between prescriptions and the medicine-management module.

MEDICINE MANAGEMENT MODULE

16 The medicine management module maintains the patient's medication records and enables the addition, modification, viewing, and removal of medicines. Each medicine record can be updated with information such as the quantity, dosage, frequency, instructions, and scheduled time. Similarly, the information extracted by the OCR module can also be converted into medication records. Therefore, the module presents a central component providing a patient's current medicines and managing multiple medicine records in an organised manner.

15 MEDICATION SCHEDULING AND REMINDER MODULE

This module, will be provided with the decision of the model, along with the explanation for the decision. The human user, must vote if the model's reasoning is valid or not. Based on the majority voting mechanism, the feedback module will be triggered. This module ensures the adaptability of the model, as certain context will be "toxic" for one period and not in other timelines. Through reinforcement such changes in the model, is updated timely, for a more stable and reliable system.

8 5 IV. RESULT AND ANALYSIS

3 A part of the proposed Intelligent healthcare Management system implementation as well as the analysis of its' various modules application have been carried out and visualized below. The system was tested on the array of healthcare-related services, including, but not limited to, Appointment Scheduling, Prescription OCR, Medicine Management, Medication Reminders, Medication Adherence Analysis, Drug-Interaction Detection, Refill Prediction. The figures below illustrate how exactly the system was employed to carry out each of the described services and support

an ongoing healthcare experience as a single process featuring multiple stages rather than a set of separated and individual applications.

Figure 4.1 below represents the Appointment Scheduling process within the proposed system. It visualizes the management of healthcare services by the patients through scheduling their medical appointments. Figure 4.2 below illustrates the functionality of the Prescription OCR within the scope of the system, representing the process of medication data recognition and extraction. In particular, the figure depicts an example of medication record production based on information pulled from the prescription. Figure 4.3 below is demonstrating the Medicine Management and Medication Reminders process and functionality.

It visualizes the medication list with corresponding dose timing and reminders that can help patients in managing their medication consumption and adherence to the set schedule. Figure 4.4 below depicts the Medication Adherence analysis process, which is utilized by the system to keep track of medication intake. It includes the medication record based on taken, missed, or snoozed doses and adherence analytics over a specific period of time. Figure 4.5 below represents the drug-interaction and refill functionality of the proposed intelligent healthcare management concept. It visualizes the warning mechanism and medicines that might require a refill. Thus these figures depict the overall healthcare process with integrated appointment scheduling, prescription digitization, medicine management, reminders, medication adherence and refill management employed to illustrate the concept of a continuous healthcare experience with ongoing patient engagement as defined by the project's scope.

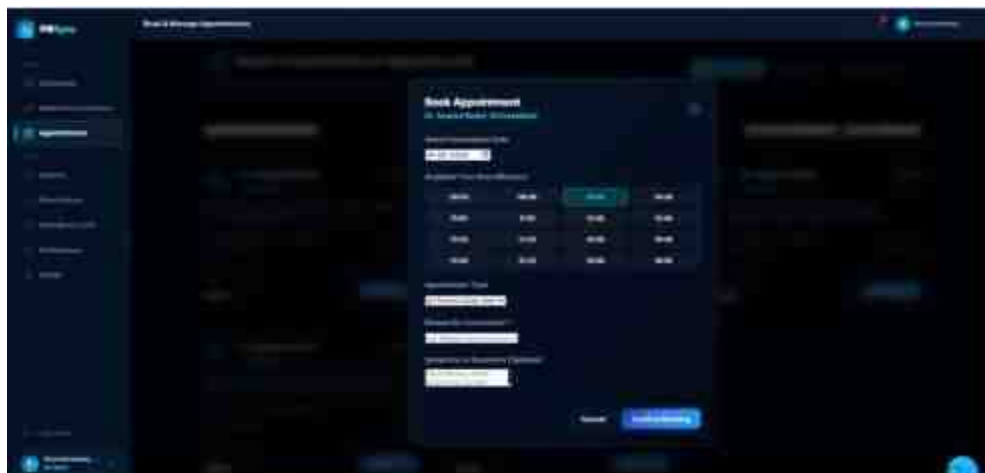


Figure 4.1. Appointment scheduling screen



Figure 4.2. OCR result

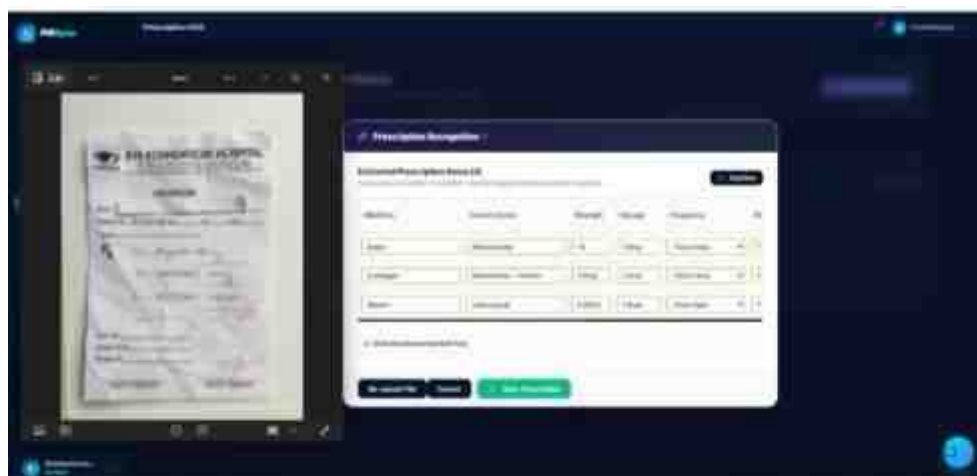


Figure 4.3. Medicine + reminder dashboard



Figure 4.4. Adherence analytics

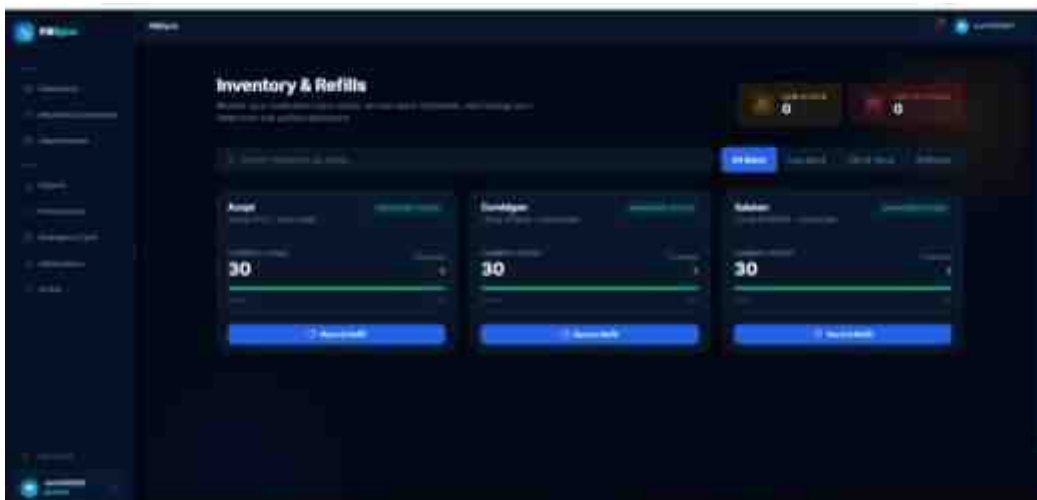


Figure 4.5. Drug interaction + refill result

V. CONCLUSION

In this paper, we have described the problem statement and objectives of the proposed Intelligent Healthcare Management system and created an integrated platform that addresses the challenges of appointment scheduling and medication management. The system is developed to provide a united solution that consists of AI-based appointment scheduling, OCR-based prescription digitisation, medicine management, medication reminders, drug-interaction detection, adherence monitoring, and refill prediction. By unifying appointment scheduling and prescription and post-

consultation medication management into one system, the proposed system aims to offer a more streamlined, continuous, and patient-centric healthcare experience. The integration of multiple healthcare modules provides a comprehensive platform for medication organisation, adherence support, and reduced manual healthcare management.

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